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B.Tech. Degree IV Semester Special Supplementary Examination September 2014

CS 1404 AUTOMATA LANGUAGES AND COMPUTATIONS (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

(8 × 5 = 40)

- I. (a) Design a finite automata which accepts the language
 $L = \{w \in \{0,1\}^* \mid \text{second symbol of } w \text{ is } 0 \text{ and fourth input is } 1.\}$
- (b) If r is a regular expression, then prove that, there exists an NFA with ϵ -transitions, that accepts $L(r)$.
- (c) State the pumping lemma for context free languages.
- (d) Write equivalent grammar in Chomsky normal form for the context free grammar
 $G = (\{S, A, B\}, \{a, b\}, P, S)$ that has productions
 $S \rightarrow bA \mid aB$
 $A \rightarrow bAA \mid aS \mid a$
 $B \rightarrow aBB \mid bS \mid b$
- (e) Explain the turing machine model with a neat diagram.
- (f) Define a multi tape turing machine. Is it possible for a multitape turing machine to accept a language L , if it is accepted by a single tape turing machine? Explain.
- (g) What are the properties of recursive and recursively enumerable languages?
- (h) Explain Chomsky hierarchy of languages.

PART B

(4 × 15 = 60)

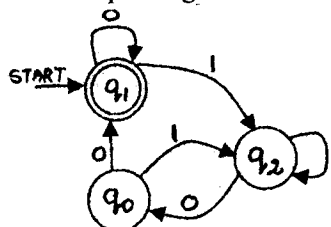
- II. (a) Construct the DFA equivalent to the NFA $M = (\{p, q, r, s\}, \{0,1\}, \delta, p, \{s\})$ where δ is given by (8)

	0	1
p	p, q	p
q	r	r
r	s	-
s	s	s

- (b) Prove the equivalence of NFA and DFA. (7)

OR

- III. (a) Construct the regular expression corresponding to the state diagram below. (8)



- (b) Differentiate between Moore and Mealy machine. (7)

(P.T.O.)

IV. (a) Design a CFG for the language $L = \{0^n 1^n \mid n \geq 0\} \cup \{1^n 0^n \mid n \geq 0\}$ (8)

(b) Using pumping lemma prove $L = \{a^i b^j c^i d^j \mid i \geq 1 \text{ and } j \geq 1\}$ is not context free. (7)

OR

V. (a) Define a PDA. Design a PDA to accept the language $L = \{a^n b^{2n}; n > 0\}$. (10)

(b) Write the CYK algorithm. (5)

VI. Design a turing machine that can compute proper subtraction that is, (15)

$m - n = m - n, \text{ if } m \geq n$
 $= 0, \text{ otherwise}$

OR

VII. (a) Explain how a turing machine can simulate subroutines. (7)

(b) Write short notes on (8)

(i) Multitape TM (ii) Multidimensional TM

VIII. (a) State the halting problem of turing machine and prove that it is undecidable. (10)

(b) Distinguish between decidable and undecidable problems. (5)

OR

IX. (a) Prove the equivalence of regular grammars and finite automata. (8)

(b) Construct left linear and right linear grammars for the language $(0+1)^* 00(0+1)^*$ (7)
